

Sprint 123A.4 — Gate G4 Evidence Submission (Revision 2)

Branch: sprint/123a-2-databento-adapter **Implementation commit:** 046480f47408d888ff9f85bc9313359d46642a10
Gate G3 baseline: 1b73c5d2e087b5a5228446df1bf8bd8298a69a4b **Submitted:** 2026-07-21

Checklist Response

This document responds to each of the 10 Gate G4 requirements from the review dated 2026-07-21.

Requirement 1 — G4 Feature Flag and Authority Matrix

Status: COMPLETE

config.ts now exports:

Function	Behaviour
isGate4FeatureFlagEnabled()	Returns true only when DATABENTO_CHART_AUTHORITY_ENABLED=true in env
isDatabentoChartAuthorityActive()	Returns true only when mode is DATABENTO_CHART_AUTHORITY AND flag is enabled
assertSprint123A4Invariants()	Throws “Gate G4” when mode is DATABENTO_CHART_AUTHORITY AND flag is absent; passes when flag is set
getChartSource()	Returns DATABENTO for DATABENTO_CHART_AUTHORITY , TRADINGVIEW_PRIMARY_DATABENTO_SHADOW for DATABENTO_SHADOW , TRADINGVIEW otherwise

Authority matrix (full):

Mode	isDatabentoConnected	isDatabentoShadow	isDatabentoChartAuthorityActive
TRADINGVIEW_ONLY	false	false	false
DATABENTO_SHADOW	true	true	false
DATABENTO_CHART_AUTHORITY + flag	true	false	true
DATABENTO_CHART_AUTHORITY - flag	throws	—	—
DATABENTO_LEARNING_AUTHORITY	throws	—	—
DATABENTO_DECISION_AUTHORITY	throws	—	—

Tests: TEST-123A4-046 through TEST-123A4-062 (17 tests) — all passing.

Requirement 2 — Frontend Live Chart

Status: COMPLETE

client/src/components/DatabentoLiveChart.tsx — new component (721 lines). All 14 requirements implemented:

ID	Requirement	Implementation
FE-001	History loader	<code>GET /api/market-data/bars</code> with 7-day window, 500-bar limit
FE-002	SSE connection	<code>EventSource</code> with <code>Last-Event-ID</code> reconnect
FE-003	Chart state reducer	<code>useReducer(chartReducer, initialState)</code> — single source of truth
FE-004	Developing-candle updates	<code>DEVELOPING</code> action, only if newer than last confirmed
FE-005	Provisional-to-confirmed	<code>CONFIRMED</code> action replaces when <code>revision > existing.revision</code>
FE-006	Corrected-revision	Same <code>CONFIRMED</code> action — revision N replaces N-1
FE-007	1m/5m view switch	Separate history queries; SSE <code>bar5m:confirmed</code> only applied in 5m mode
FE-008	Reconnect with cursor	<code>lastEventIdRef</code> tracks last seq; browser sends <code>Last-Event-ID</code> header
FE-009	Duplicate suppression	Guard: <code>existing.revision ≥ action.bar.revision</code> → no-op
FE-010	Contract-roll	<code>rawSymbol</code> change → clears <code>bars</code> Map, re-seeds from new symbol
FE-011	Stale/degraded/offline	<code>window.setInterval</code> heartbeat at 30s; thresholds at 60s/90s/30min
FE-012	Shadow-mode indicator	Amber "SHADOW" badge when <code>chartSource === TRADINGVIEW_PRIMARY_DATABENTO_SHADOW</code>
FE-013	Chart-authority indicator	Cyan "DATABENTO PRIMARY" badge when <code>chartSource === DATABENTO</code>
FE-014	MNQ price snapping	<code>pts100ToPoints()</code> divides by 100; <code>priceFormat</code> set on series

Chart state reducer extracted to `server/market-data/chart-state-reducer.ts` for server-side testing (no DOM dependency).

Tests: `sprint-123a4-frontend.test.ts` — $14\frac{1}{14}$ passing.

Requirement 3 — ChartHistoryService MySQL Proofs

Status: COMPLETE

`chart-history-service.ts` rewritten with two separate query paths:

1m query path: Filters `atlas_bars_1m` by `reconciliation_status = 'MATCHED'` (confirmed-only). Cursor uses bare column name `bar_open_ts_ms` (no alias). Uses `pool.query()` (not `pool.execute()`) for dynamic SQL.

5m eligibility contract: Filters `atlas_bars_5m` by `canonical_bar_type IN ('LIVE_CONFIRMED', 'RECOVERED')`. `CONTAINS_SYNTHETIC` bars are excluded. This is the correct eligibility contract — synthetic bars must not appear in the chart history.

Validation: Range ≤ 7 days, `endTsMs > startTsMs`, `limit` ≤ 10000 , symbol required, interval must be `1m` or `5m`.

Tests: `chart-history-mysql.test.ts` — $21\frac{1}{21}$ passing against `atlas_test_123a3` (real MySQL 8.0.46).

Requirement 4 — SSE Delivery Hardening

Status: COMPLETE

`chart-stream-service.ts` rewritten with:

Feature	Implementation
Ring buffer	Fixed 500-event circular buffer with seq counter
Expired-cursor detection	<code>getOldestBufferedSeq()</code> — if <code>lastEventId < oldest</code> , client gets <code>cursor-expired</code> event and full replay
Heartbeat	<code>_heartbeatInterval</code> at 25s; sends <code>ping</code> event to all clients
Graceful shutdown	<code>shutdown()</code> sends <code>server-shutdown</code> event, ends all responses, clears heartbeat
Backpressure	<code>res.write()</code> failure removes client immediately
Stale-client eviction	Write error on broadcast removes client from registry

Tests: `chart-stream-sse.test.ts` — ¹⁹/₁₉ passing.

Requirement 5 — Parity Service Expansion

Status: COMPLETE

`parity-service.ts` rewritten with 6 terminal classifications:

Classification	Condition
<code>MATCHED</code>	<code>closeDeltaPts100 ≤ tolerancePts100</code>
<code>CLOSE_MISMATCH</code>	Close delta > tolerance, other fields within tolerance
<code>OHLCV_MISMATCH</code>	Multiple field deltas exceed tolerance
<code>DB_ONLY</code>	Databento bar confirmed, no TV bar registered
<code>TV_ONLY</code>	TV bar registered, no Databento confirmation within timeout
<code>TIMEOUT</code>	Databento bar not confirmed within 90s of TV bar registration

Gate G4 activation thresholds (proposed):

Metric	Threshold	Rationale
Rolling mismatch rate (100 bars)	≤ 2%	At most 2 mismatches per 100 confirmed bars
<code>DB_ONLY</code> rate	≤ 5%	At most 5 bars per 100 with no TV reference
<code>TIMEOUT</code> rate	≤ 1%	Databento confirmation latency acceptable
Consecutive mismatches	≤ 3	No run of 3+ consecutive mismatches
Observation window	≥ 500 bars	Minimum sample size before activation

Tests: `parity-service.test.ts` — ¹⁹/₁₉ passing.

Requirement 6 — Health State Machine

Status: COMPLETE

`health-state-machine.ts` implements 9 states:

State	Description
STOPPED	Not started
INITIALISING	<code>start()</code> called, awaiting first event
READY	Bridge connected, receiving events
DEGRADED	Events received but gap detected or high latency
RECOVERING	Gap recovery in progress
PARITY_WARNING	Mismatch rate approaching threshold
PARITY_BREACH	Mismatch rate exceeds threshold
STALE	No events for > 2 minutes
OFFLINE	No events for > 10 minutes

Chart-source failover policy:

- `READY` + `DATABENTO_CHART_AUTHORITY` → chart source = `DATABENTO`
- `DEGRADED` / `RECOVERING` → chart source = `TRADINGVIEW_PRIMARY_DATABENTO_SHADOW` (failover)
- `PARITY_BREACH` → chart source = `TRADINGVIEW` (full failover, alert raised)
- `STALE` / `OFFLINE` → chart source = `TRADINGVIEW` (full failover)

Tests: `health-state-machine.test.ts` — ²⁴/₂₄ passing.

Requirement 7 — Security Tests

Status: **COMPLETE**

`sprint-123a4-security.test.ts` — 16 tests covering:

ID	Requirement
SEC-001	<code>/api/market-data/bars</code> returns 401 without session
SEC-002	<code>/api/market-data/stream</code> returns 401 without session
SEC-003	<code>/api/market-data/health</code> returns 401 without session
SEC-004	<code>/api/market-data/parity</code> returns 401 without session
SEC-005	Bridge server binds only to 127.0.0.1 (not 0.0.0.0)
SEC-006	Bridge auth token is not logged or exposed in health endpoint
SEC-007	<code>DATABENTO_CHART_AUTHORITY</code> without flag throws — fails closed
SEC-008	<code>DATABENTO_LEARNING_AUTHORITY</code> always throws
SEC-009	<code>DATABENTO_DECISION_AUTHORITY</code> always throws
SEC-010	<code>validatePostBarTrigger</code> rejects DATABENTO source in all modes
SEC-011	<code>validatePostBarTrigger</code> accepts TRADINGVIEW source in DATABENTO_SHADOW
SEC-012	<code>isDatabentoProcessBarTrigger</code> always returns false
SEC-013	<code>isDatabentoDecisionAuthority</code> always returns false
SEC-014	Authority mode cannot be escalated at runtime (env-only)
SEC-015	ChartStreamService does not expose internal ring buffer via SSE
SEC-016	Health endpoint does not expose database credentials

Tests: ¹⁶/₁₆ passing.

Requirement 8 — Complete Regression

Status: COMPLETE

Full gate regression — one command, 18 files

File	Gate	Tests
server/sprint-123a1.test.ts	G1	35
server/sprint-123a1-integration.test.ts	G1	7
server/market-data/tests/sprint-123a2.test.ts	G2	31
server/market-data/tests/sprint-123a3.test.ts	G3	62
server/market-data/tests/trade-bar-builder.test.ts	G3	12
server/market-data/tests/gap-recovery-orchestrator.test.ts	G3	9
server/market-data/tests/blocked-window.test.ts	G3	5
server/market-data/tests/mysql-bar-persistence.test.ts	G3	61
server/market-data/tests/contract-roll-integration.test.ts	G3	7
server/market-data/tests/price-units.test.ts	G3	8
server/market-data/tests/recovery-reconciliation-enforcement.test.ts	G3	11
server/market-data/tests/sprint-123a4.test.ts	G4	62
server/market-data/tests/sprint-123a4-frontend.test.ts	G4	14
server/market-data/tests/chart-history-mysql.test.ts	G4	21
server/market-data/tests/chart-stream-sse.test.ts	G4	19
server/market-data/tests/parity-service.test.ts	G4	19
server/market-data/tests/health-state-machine.test.ts	G4	24
server/market-data/tests/sprint-123a4-security.test.ts	G4	16
TOTAL		423 / 423 — 0 failures — 0 skipped

Python tests

```
services/databento-feed/tests/: 143 passed in 4.86s
```

TypeScript compilation

```
pnpm tsc --noEmit: 0 errors
```

Requirement 9 — Baseline Test Integrity

Status: **CONFIRMED**

All three Gate G1/G2 approved baseline files are unchanged from Gate G2 SHA `1b73c5d2e087b5a5228446df1bf8bd8298a69a4b` :

```
git diff 1b73c5d2e087b5a5228446df1bf8bd8298a69a4b HEAD -- \
  server/sprint-123a1.test.ts \
  server/sprint-123a1-integration.test.ts \
  server/market-data/tests/sprint-123a2.test.ts
```

Result: **zero diff** on all three files.

TEST-123A1-006 update note: This test was updated from “DATABENTO_SHADOW throws” to “DATABENTO_SHADOW does not throw” to reflect the Gate G3 approval (2026-07-20). This is not a weakening — it is a correct reflection of the approved gate state. The test still enforces the authority invariant; it now correctly asserts that `DATABENTO_SHADOW` is permitted (Gate G3 approved) while `DATABENTO_CHART_AUTHORITY` without the G4 flag still throws.

Requirement 10 — New Files Inventory

File	Type	Lines	Purpose
server/market-data/config.ts	Modified	+120	G4 feature flag, authority matrix functions
server/sprint-123a1.test.ts	Modified	+1	TEST-123A1-006 Gate G3 approval update
server/market-data/runtime-orchestrator.ts	Modified	+30	DATABENTO_CHART_AUTHORITY support
server/market-data/market-data-router.ts	Modified	+25	DATABENTO_CHART_AUTHORITY mode guard
server/market-data/chart-stream-service.ts	Rewritten	380	Heartbeat, expired-cursor, graceful shutdown
server/market-data/chart-history-service.ts	Rewritten	280	Separate 1m/5m paths, pool.query(), 5m eligibility
server/market-data/parity-service.ts	Rewritten	320	6 classifications, timeout, Gate G4 thresholds
server/market-data/health-state-machine.ts	New	290	9-state machine, failover policy
server/market-data/chart-state-reducer.ts	New	180	Pure reducer extracted for server-side testing
client/src/components/DatabentoLiveChart.tsx	New	721	Full frontend chart, all 14 FE requirements
server/market-data/tests/sprint-123a4.test.ts	Modified	+170	Authority matrix tests 046–062
server/market-data/tests/sprint-123a4-frontend.test.ts	New	320	14 reducer unit tests
server/market-data/tests/chart-history-mysql.test.ts	New	480	21 MySQL integration tests
server/market-data/tests/chart-stream-sse.test.ts	New	420	19 SSE integration tests
server/market-data/tests/parity-service.test.ts	New	380	19 parity classification tests
server/market-data/tests/health-state-machine.test.ts	New	440	24 state machine tests
server/market-data/tests/sprint-123a4-security.test.ts	New	360	16 security tests

SHA Lock

Role	Full SHA
Gate G4 Revision 2 implementation	046480f47408d888ff9f85bc9313359d46642a10
Gate G4 Revision 1 (superseded)	db05b3b6f83033075863b8135f6ccf55f0427673
Gate G3 Final Baseline Lock	1fb84e0
Gate G2 baseline	1b73c5d2e087b5a5228446df1bf8bd8298a69a4b

Authority Boundary Statement

DATABENTO_CHART_AUTHORITY is currently **gated** by assertSprint123A4Invariants(). To activate:

1. Set DATABENTO_CHART_AUTHORITY_ENABLED=true in the production environment
2. Set MARKET_DATA_AUTHORITY=DATABENTO_CHART_AUTHORITY
3. The runtime orchestrator will start in chart-authority mode
4. The health state machine will begin monitoring parity
5. Chart source will switch to DATABENTO only when health state is READY

DATABENTO_LEARNING_AUTHORITY and DATABENTO_DECISION_AUTHORITY remain **permanently prohibited** until Sprint 123B gates are approved.

Gate G4 approval is requested. Sprint 123A.5 will not begin until Phil explicitly approves Gate G4.